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Roku JavaScript Library

Roku's JavaScript Library enables keyword-driven testing of apps. The library resides in a JavaScript class that has methods that map directly to keyword names. The keywords take the same arguments as the methods implementing them. The keywords report failures with exceptions, create logs by writing to standard output, and return values using the `return` statement.

Instantiating the library

To create an instance of the Roku JavaScript Library, provide the following three arguments:

Argument	Description
ip	The IP address of the device to be used for testing.
timeout	The amount of time (in milliseconds) that commands are allowed to run.
pressDelay	The amount of time (in milliseconds) between keypress commands. This argument works with the sendKeys command.

The following example demonstrates how to instantiate the Roku JavaScript Library:

```
library = new rokuLibrary.Library("192.168.2.121", 20000, 2000);
```

The following example demonstrates how to instantiate the Roku JavaScript Library and runs test case with Mocha and Chai:

```
//import library
const rokuLibrary = require("../library/rokuLibrary");
const expect = require("chai").expect;
const {
   spawn
} = require('child_process');

//start instance of webDriver
const childProcess = spawn('D:/projects/go/webDriver/src/main.exe');
let library;

//test suite
describe('test_3-Grid', () => {
   //this method is executed before all test cases
   before(() => {
      //create instance of jsLibrary
      library = new rokuLibrary.Library("192.168.2.121", 20000, 2000);
   });

   //test case
   it('should launch the app', async function() {
      this.timeout(25000);
      await library.sideLoad("../channels/3_Grid.zip", "rokudev", "your_channel_password");
      await library.verifyIsChannelLoaded('dev');
   });

   //this method executed after all tests
   after(async () => {
      //close session
      await library.close();
      //exit webDriver
      childProcess.kill();
   });
});
```

Keywords

The Roku's JavaScript Library includes the following keywords:

- Sideload (*available since release 2.0*)
- Launch the app
- Input deep linking data (*available since release 2.0*)
- Get apps
- Send key
- Send keys

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- Send word
- Mark timer (*available since release 2.0*)
- Get timer (*available since release 2.0*)
- Verify is playback started
- Verify is screen loaded
- Get child nodes (*available since release 2.1*)
- Get element
- Get elements
- Get focused element
- Verify is app loaded
- Get current app info
- Get device info
- Get player info
- Verify is channel exist
- Set timeout
- Set press delay
- Get attribute

A keyword will fail if its respective WebDriver endpoint returns a 4xx or 500 error.

sideLoad

(*available since release 2.0*)

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Keyword	Argument	Description	Example
sideLoad	• channel: The file path of a zipped package file. • username: Enter rokudev, which is the user name for the Development Application Installer. • password: The password for accessing the Development Application Installer on your Roku device.	Sideloads an app that has been packaged into a zip file. If the sideLoad command fails, sideload (/docs/developer-program/getting-started/developer-setup.md#sideloading-channels) the app to be tested and use the launchTheChannel command.	`await library.sideLoad("channel.zip`

launchTheChannel

Keyword	Argument	Description	Example
launchTheChannel	• channel_code: The ID of the app to be launched. • contentId: The contentId (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentType to execute deep linking tests. • mediaType: The mediaType (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentId to execute deep linking tests.	Launches the app corresponding to the specified app ID.	`await lib`

inputDeepLinkingData

(*available since release 2.0*)

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Keyword	Argument	Description	Exam
inputDeepLinkingData	• channelId: The ID of the app to be launched. • contentId: The contentId (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentType to execute deep linking tests. • mediaType: The mediaType (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentId to execute deep linking tests.	Launches the app corresponding to the specified app ID.	awa

getApps

Keyword	Description	Example
getApps	Returns a list of installed apps as an array of objects. Each app object contains the following fields: • title • id • type • version • subtype	expect(apps[0].ID).to.equal('some_id') const apps = await library.getApps()

sendKey

Keyword	Arguments	Description	Example
sendKey	• key_press: The key to be pressed and released, which may be one of the following: "up", "down", "right", "left", "back, "select", "instantreplay", "play", "stop", "rev", "fwd", and "info". • delay: The delay (in seconds) before the keypresses are executed. This argument is optional, and it defaults to 2 seconds if not specified.	Simulates the press and release of the specified key.	`await library.sendKey('select', 4)`

sendKeys

Keyword	Arguments	Description	Example
sendKeys	• sequence: An array containing the sequence of keys to be pressed and released (for example, down, down, down, down, select). • delay: The delay (in seconds) before the keypresses are executed. This argument is optional, and it defaults to 2 seconds if not specified.	Simulates the sequence of keypresses and releases.	`await library.sendKeys(['down', 'down', 'down',`

sendWord

Keyword	Arguments	Description	Example
sendWord	• word: The specified word to be entered. • delay: The delay (in seconds) before the entry of each letter in the specified word. This argument is optional, and it defaults to 2 seconds if not specified.	Simulates the press and release of each letter in a word.	`await library.sendWord('user')`

markTimer

(*available since release 2.0*)

Keyword	Description	Example
markTimer	Starts the timer.	`library.markTimer();`

getTimer

(*available since release 2.0*)

Keyword	Description	Example

getTimer	Returns the number of milliseconds elapsed since the timer was last started.	`let res = await library.verifyIsPlaybackStarted(25, 1);`
		`expect(res).to.equal(true);`
		`let time = library.getTimer();`
		`expect(14000).greaterThan(time);`

verifyIsPlaybackStarted

Keyword	Arguments	Description	Example
verifyIsPlaybackStarted	• retries: The number of requests that can be made before returning false. This argument is optional, and it defaults to 10 if not specified. • delay: The delay (in seconds) between retries. This argument is optional, and it defaults to 1 second if not specified.	Verify playback has started on the Roku media player. This keyword fails if player state is not "play".	`verifyIsPlaybackStarte`

verifyIsScreenLoaded

Keyword	Arguments	Description	Example
Verify is screen loaded	• data: An object with locators for elementData and parentData (parentData is optional). See the WebDriver element command (/docs/developer-program/dev-tools/automated-channel-testing/web-driver.md#POST-v1/session/:sessionId/elements) command for more information. • retries: The number of requests that can be made before returning false. This argument is optional, and it defaults to 10 if not specified. • delay: The delay (in seconds) between retries. This argument is optional, and it defaults to 1 second if not specified.	Verify that the screen is loaded based on the provided element data.	`verifyIsScreenLoade`

getChildNodes

Keyword	Arguments	Description	Example

getChildNodes	• parentNode: The parent node for which the child nodes are to be retrieved.	Retrieves the child component of the specified node.	`const rowList = await library.getE`
			`const searchData = [{"using": "tag`
			`const result = library.getChildNod`

- locator (optional): An array containing search criteria for the child nodes to be retrieved. The locator has the following syntax:

```
"using" ("attr", "tag", "text") : specify locator type
"value": tag or attribute value
"attribute"(only for "attr" type): specify attribute
```

getElement

Keyword	Arguments	Description	Example
Get element	• data: An object with locators for elementData and parentData (parentData is optional). See the WebDriver element command (/docs/developer-program/dev-tools/automated-channel-testing/web-driver.md#POST-v1/session/:sessionId/elements) for more information. • delay: The delay (in seconds) between retries. This argument is optional, and it defaults to 1 second if not specified.	Searches for an element on the page based on the specified locator starting from the screen root. Returns information on the first matching element.	const element = awai const poster = library expect(poster).to.equ

getElements

Keyword	Arguments	Description	Example
getElements	• data: An object with locators for elementData and parentData (parentData is optional). See the WebDriver element command (/docs/developer-program/dev-tools/automated-channel-testing/web-driver.md#POST-v1/session/:sessionId/elements) for more information. • delay: The delay (in seconds) between retries. This argument is optional, and it defaults to 1 second if not specified.	Searches for elements on the page based on the specified locators starting from the screen root. Returns information on the matching elements.	const elements= const poster = lib expect(poster).to

getFocusedElement

Keyword	Description	Example
getFocusedElement	Return the element on the screen that currently has focus. See the WebDriver active element command (/docs/developer-program/dev-tools/automated-channel-testing/web-driver.md#get-v1/session/:sessionId/element/active) for more information.	const element = await library.ge const poster = library.getAttribu expect(poster).to.equal('poster.p

verifyIsChannelLoaded

Keyword	Arguments	Description	Example
verifyIsChannelLoaded	• id: The ID of the app to be launched. Use `dev` to verify a sideloaded app. • retries: The number of requests that can be made before returning false. This argument is optional, and it defaults to 10 if not specified. • delay: The delay (in seconds) between retries. This argument is optional, and it defaults to 1 second if not specified.	Verify that the specified app has been launched. This keyword fails if the provided app ID does not match a valid app.	`await library.verifyIsCha`

###getCurrentChannelInfo

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Keyword	Description	Example
getCurrentChannelInfo	Returns an object containing information about the app currently loaded. This object has the following fields:	const data = av expect(data.Po

Key	Type	Description
sessionId	string	The advertising ID of the device
status	int	A status code summarizing the result of the command.
value	array	
value[i].Title	string	The title of the app.
value[i].ID	string	The ID of the app.
value[i].Version	string	The build version of the app.
value[i].Subtype	string	"ndka"/"rsga"
value[i].Type	string	"menu"/"appl"

getDeviceInfo

Keyword	Description	Example
getDeviceInfo	Returns an object containing the information about the device. This object has the following fields:	const info = await libr expect(info.language)

Key	Type	Description
sessionId	string	The advertisement ID of the device.
status	int	A status code summarizing the result of the command.
value	object	
value.vendorName	string	The vendor of the device.
value.modelName	string	The model of the device.
value.language	string	The language of the device.
value.country	string	The country of the device.
value.ip	string	The IP address of the device.
value.timeout	int	The specified timeout for WebDriver client requests.
value.pressDelay	int	The specified delay between key presses.

getPlayerInfo

Keyword	Description	Ex
getPlayerInfo	Returns an object containing information about the Roku media player. This object has the following fields:	

Key	Type	Description
sessionId	string	The advertising ID of the device
status	int	A status code summarizing the result of the command
value	object	
value.error	string	Indicates whether there was a playback error. If no error occurred, this is set to "false"
value.state	string	Indicates the current playback state ("play", "pause", "resume", and so on)
value.format	object	The format element contains the following attributes: *audio*, *caption*, *container*, *drm*, *video*, and *res*.
value.format.audio	string	The audio compression method ("aac", "aac_adts", and so on.)
value.format.caption	string	The closed caption format ("608_708", for example). This value is set to "none" if there are no captions.

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Keyword	Description	Ex

Key	Type	Description
value.format.container	string	The container format ("hls", for example)
value.format.drm	string	The encoding type. If no encoding is used, this us set to "none".
value.format.video	string	The format of the currently playing video stream ("mpeg4-15", for example)
value.format.res	string	The resolution of the currently playing video stream ("1280X720", for example).
value.buffering	object	The buffering element contains the following attributes: *current*, *max*, *target*.
value.buffering.current	string	The current buffering speed (in kbps).
value.buffering.max	string	The maximum possible buffering speed (in kbps).
value.buffering.target	string	The target buffering speed (in kbps).
value.newStream	object	The newStream element contains the following attribute: *speed*.

Keyword	Description	Ex

Key	Type	Description
value.newStream.speed	string	The current playback speed (in bps)
value.position	string	The time of the current position in the stream, expressed as the elapsed time (in ms) since the start of stream or UTC time, depending on the content.
value.duration	string	The duration of the video being played (in seconds). This becomes valid when playback begins and may change if the video is dynamic content, such as a live event.
value.isLive	string	A flag indicating whether the video being played is a live stream.
value.runtime	string	The runtime of the video being played (in seconds).

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Keyword	Description	Ex

Key	Type	Description
value.streamSegment	object	The streamSegment attribute contains Information about the video segment that is currently streaming. This is only meaningful for segmented video transports, such as DASH and HLS This element contains the following attributes: *bitrate*, *mediaSequence*, *segmentType*, and *time*.
value.streamSegment.bitrate	string	The bitrate of the video segment (in bps).
value.streamSegment.mediaSequence	string	The HLS media sequence ID of the segment in the video.
value.streamSegment.segmentType	string	The type of data in the segment, which may be one of the following values: "audio", "video", "captions", "mux".
value.streamSegment.time	string	The chunk start time.

verifyIsChannelExist

Keyword	Arguments	Description	Example

verifyIsChannelExist	• apps: An array containing currently installed on the device.
	• id: The ID of the app to be verified. Use `dev` to verify a sideloaded app.
		Verifies the specified app is installed on the device. This keyword fails if the apps array does not contain the app specified in the id argument.	`const apps = await library.g`
			`const res = library.verifyIsCl`
			`expect(res).equal(true);`

setTimeout

Keyword	Arguments	Description	Example
setTimeout	timeout: The amount of time (in milliseconds) that Web driver client requests are allowed to run.	Sets the timeout for Web driver client requests.	`await library.setTimeout(10000)`

setDelay

Keyword	Arguments	Description	Example
setDelay	delay: The interval (in milliseconds) to be used between key presses.	Sets the delay between key presses. This keyword works with the Send keys keyword.	`await library.setDelay(1000)`

getAttribute

Keyword	Arguments	Description	Example
getAttribute	• element: An object that contains element information (attributes, child nodes). • attr: The name of the attribute to be retrieved.	Get attribute value. This keyword fails if an element does not contain the specified attribute.	const elements= await library.getEl const poster = library.getAttribute(e expect(poster).to.equal('poster.png'

Sample test cases

The Roku automated app testing repository (https://github.com/rokudev/automated-channel-testing) includes a set of sample JavaScript test cases that can be executed on their corresponding SceneGraph Developer Extensions (SGDEX) sample apps (https://github.com/rokudev/SceneGraphDeveloperExtensions/tree/master/samples). For example, you can execute the SGDEX GridView test case (test_3_Grid.js), which will sideload the corresponding sample app (3_Grid) on your device, and then view the test output. You can reference these samples when developing test scripts for the automated testing of your development apps.

> Before running a sample test case, you need to update the sideload command in the test case with the Roku device password.

The test_basic.js sample (https://github.com/rokudev/automated-channel-testing/blob/master/jsLibrary/tests/test_basic.js) demonstrates how to create a simple test case that checks whether a user is authenticated before playing content using the Roku JavaScript Library:

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```
const rokuLibrary = require("../library/rokuLibrary");
const expect = require("chai").expect;
const {
    spawn
} = require('child_process');
const childProcess = spawn('D:/projects/go/webDriver/src/main.exe');
let library;
describe('App should be launched', () => {
    before(() => {
        library = newrokuLibrary.Library("192.168.1.64");
    });

    it('Check if app exists on the device', asyncfunction() {
        this.timeout(5000);
        const apps = await library.getApps();
        const res = library.verifyIsChannelExist(apps, 'dev');
        expect(res).equal(true);
    });

    it('should launch the app', asyncfunction() {
        this.timeout(5000);
        await library.launchTheChannel('dev');
        await library.verifyIsChannelLoaded('dev');
    });

    after(async () => {
        await library.close();
        childProcess.kill();
    });

});
```

Viewing the test case report and log

After you run a test case that uses the Roku JavaScript Library, you can view the generated report and log files in the specified output directory. The report summarizes the test case and provides statistics on the percentage of individual tests that passed/failed. The log details the success/failure of the individual keywords used in each test case.

Previous

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